

Adaptive Shell Weighting and the Nonnegative Coherence Wall in a Literal Twin-Prime Dynamics

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Abstract

TPC-289 found a finite late-shell block whose physical prime-component vectors have positive and quantitatively large cross-Gram terms, together with three early sign-flip pairs. We study the natural adaptive response: can nonnegative shell weights create energy decay? For vectors g_q with Gram matrix G and diagonal d_q , we introduce

$$R(w) = \frac{\left\| \sum_q w_q g_q \right\|^2}{\sum_q w_q^2 d_q}, \quad \kappa(w) = \frac{(\sum_q w_q)^2}{\sum_q w_q^2}.$$

We prove exactly that nonnegative weights cannot give $R(w) < 1$ when all off-diagonal Gram entries are nonnegative. Under a positive coherence floor and diagonal balance we prove the sharper effective-support bound $R(w) \geq 1 + \eta\delta(\kappa(w) - 1)$. An exact-rational replay of the TPC-289 18-row grid tests uniform, inverse-diagonal, and linear-taper full-support policies, every equal two-prime support, and every leave-one-out support. All 54 full-support policy rows and all 18 leave-one-out minima are amplified. Exactly three subunit equal-pair witnesses occur, all in the one early sign-flip row. Thus positive diffuse adaptation remains behind the coherence wall; the only finite nonnegative escape is sparse concentration on a sign-flip pair. This is a structural obstruction, not an asymptotic weighted theorem or a twin-prime result.

1 Route position and question

We keep one literal prime-shell dynamical family throughout. TPC-288 lifted scalar cancellation to the physical output Gram and showed finite full-rank energy amplification [2]. TPC-289 then proved a conditional coherence accumulation bound and found an eight-row late positive block, but also found three negative pairs at the $(N, H, Q, z, s) = (256, 38, 27, 5, 1)$ crossover. The inherited coherence phase diagram is recorded in TPC-289 [1].

The phrase “adaptive weighting” hides two different possibilities. A diffuse nonnegative reweighting preserves the physical sign cone, whereas a sparse rule may select an exceptional negative pair. The purpose of this paper is to separate these mechanisms with an exact weighted Gram identity and a finite certificate. We do not alter the source, kernel, deletion rule, or prime shell inherited from TPC-289.

2 Frozen physical model

Let $I = I_N$ be the frozen integer interval and let β be its rational source vector. For an odd prime q , write

$$K_{H,s}(h) = \frac{H^{2s}}{(H^2 + h^2)^s}, \quad B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

The deleted-diagonal component is

$$(A_q)_{u,t} = qK_{H,s}(u-t) \mathbf{1}_{u \neq t} B_q(u, t), \quad g_q = A_q \beta.$$

For a shell S , set

$$G_{q,r} = \langle g_q, g_r \rangle, \quad d_q = G_{q,q} = \|g_q\|^2.$$

The finite computations have $d_q > 0$. The unweighted physical ratio from the preceding release is $\left\| \sum_q g_q \right\|^2 / \sum_q d_q$.

3 Weighted Gram lemmas

Lemma 1 (Weighted identity). *For every nonzero $w \in \mathbb{R}^S$,*

$$R(w) = 1 + \frac{2 \sum_{q < r} w_q w_r G_{q,r}}{\sum_q w_q^2 d_q}. \quad (1)$$

Proof. Expand $\left\| \sum_q w_q g_q \right\|^2 = w^T G w$. The diagonal part is the positive denominator and the off-diagonal terms occur twice by symmetry. $\square$

Proposition 1 (Nonnegative coherence wall). *If $w_q \geq 0$ and $G_{q,r} \geq 0$ for every distinct pair, then $R(w) \geq 1$. If at least one pair has $w_q w_r G_{q,r} > 0$, the inequality is strict.*

Proof. Every summand in the numerator of the correction term in (1) is nonnegative, and its denominator is positive. $\square$

Proposition 2 (Effective-support accumulation). *Suppose $w_q \geq 0$, $G_{q,r} \geq \eta \sqrt{d_q d_r}$ for $q \neq r$, and $d_{\min}/d_{\max} \geq \delta$, with $\eta, \delta \geq 0$. Then*

$$R(w) \geq 1 + \eta \delta (\kappa(w) - 1), \text{quad} \kappa(w) = \frac{(\sum_q w_q)^2}{\sum_q w_q^2}. \quad (2)$$

Proof. The hypotheses give $G_{q,r} \geq \eta d_{\min}$. Hence the correction in (1) is at least

$$\frac{2\eta d_{\min} \sum_{q < r} w_q w_r}{d_{\max} \sum_q w_q^2}.$$

Use $2 \sum_{q < r} w_q w_r = (\sum_q w_q)^2 - \sum_q w_q^2$ and the diagonal ratio. $\square$

Lemma 2 (Equal-pair probe). *For weights supported equally on i, j ,*

$$R_{i,j} = 1 + \frac{2G_{i,j}}{d_i + d_j}. \quad (3)$$

Thus a negative cross term is an explicit nonnegative subunit witness.

Proof. Substitute $w_i = w_j = 1$ and all other coordinates zero in (1). $\square$

For uniform weights, $\kappa(w) = |S|$, so (2) recovers the TPC-289 lower bound. The new parameter records the cost of concentration: it is one for a single component and at most $|S|$ for nonnegative weights.

4 Finite protocol

The certificate replays eight $s = 2$ growth anchors, four $s = 1$ crossover rows, and six $s = 2$ source controls, for 18 rows and 1,380 unordered Gram pairs. All arithmetic is rational ‘Fraction’ arithmetic. The three full-support policies are

$$w_q = 1, \quad w_q = d_q^{-1}, \quad w_q = 2Q - q,$$

respectively called uniform, inverse-diagonal, and linear taper. Scaling a weight vector does not change $R(w)$. In addition, the certificate evaluates the equal-pair ratio (3) for every pair and the uniform ratio after omitting each one prime.

The positive-block thresholds are the inherited finite values $\eta = 3/5$, $\delta = 4/5$. The producer locks the TPC-289 code and result and the TPC-268 engine by normalized-LF SHA-256. A reverse-order independent replay reconstructs the rows without importing the producer. A ten-case mutation audit and normal/optimized byte-identical checks complete the local fail-closed validation.

axis	N	H	Q	s	k	$R(1)$	$R(d^{-1})$	$R(2Q - q)$
growth	128	24	9	2	3	2.124905	2.117289	1.751140
growth	192	32	16	2	5	3.443701	3.435108	2.517680
growth	256	38	27	2	7	2.622578	2.753184	2.054584
growth	512	58	60	2	13	10.992677	10.967816	8.115423
growth	512	58	90	2	17	16.439313	16.422993	12.172859
cross	256	38	27	1	7	2.166576	2.163534	1.903891
cross	512	58	70	1	15	8.900964	8.916827	6.009915
control	384	48	70	2	15	14.322934	14.297999	10.463816
control	384	52	70	2	15	14.107249	14.075231	10.228418

Table 1: Representative exact-rational rows; decimals are display only.

5 Results

All 54 full-support policy rows are amplified, including the exceptional crossover row. All 18 leave-one-out minima are also greater than one. The equal-pair scan is different: exactly three subunit records occur, namely the pairs (29, 53), (31, 53), and (41, 53) in the exceptional row. The minimum is

$$R_{29,53} \approx 0.8975237303.$$

These are nonnegative weights, but they have effective support two and do not represent the full physical shell. The other 17 rows have no subunit equal-pair probe. The eight TPC-289 strong-block rows satisfy the conditional bound (2) for every one of the three policies.

6 Interpretation and claim firewall

The exact theorem closes a tempting local escape: on an all-positive Gram block, changing positive coefficients cannot manufacture cancellation. The finite scan refines this statement rather than universalizing it. An early sign flip permits sparse nonnegative cancellation, while the full-support policies and leave-one-out tests remain amplified. In route-map language, adaptive weighting has split into a positive diffuse branch and a sparse exception branch.

The following claims are deliberately not made:

- no theorem is asserted for a growing shell or every admissible source;
- the three policies are not an optimization over all weighting rules;
- the sparse witnesses are not a full-shell L^2 saving;
- no arithmetic L^2 estimate, fixed-power credit, Gate-B conclusion, or twin-prime conclusion follows.

The next minimal attack is signed two-prime Schur cancellation, which can measure exactly how much cancellation a high-coherence pair offers and what sign cost it incurs. A complementary route is to seek a source restriction that makes the positive diffuse block stable.

7 Reproducibility

The project README, proof package, canonical JSON certificate, independent replay, stress audit, and Bridge-B checker are included with the source. The Session-named Route-A/Route-B evaluator files are absent from this checkout; no official evaluator pass is claimed. The local status is

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EXACT: weighted Gram identity; nonnegative coherence wall
CONDITIONAL: effective-support accumulation bound
FINITE: 54/54 full-support policies amplified; 18/18 drop-one amplified
FINITE: 3 sparse equal-pair subunit witnesses in one sign-flip row
OPEN: growing diffuse weighted theorem and arithmetic L2
NONE: twin-prime conclusion

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References

- [1] Liang Wang. Cross-prime gram coherence in a literal twin-prime dynamics, 2026. TPC-289 project release in the same research repository.
- [2] Liang Wang. Growing-shell gram obstruction, 2026. TPC-288 project release in the same research repository.